Experiment 17C: Mathematical Modelling and Channel-Present False-Alarm Attribution for a 5.9 GHz V2X Channel

Abstract

This report presents the theoretical basis, mathematical derivations, numerical substitutions, and interpretation of Experiment 17C. The experiment uses a Sionna RT multipath channel at a carrier frequency of 5.9 GHz, constructs a discrete-time channel response, estimates its delay profile using an OFDM reference signal, and evaluates adaptive-threshold peak detection under complex Gaussian noise. The central objective is to attribute detected peaks to (i) primary noise-free composite-profile features supported by simulated ray groups, (ii) deterministic secondary features, or (iii) detections that remain unexplained by the constructed noise-free feature catalogue. The distinction between a cell-level false-alarm probability and a profile-level false-alarm probability is explicitly discussed.

Contents

1 Experiment objective and configuration **3**

2 Multipath channel model **3**

2.1 Continuous-time representation 3

2.2 Ray grouping 4

3 Discrete-time channel construction **5**

3.1 Sampling interval and delay grid 5

3.2 Windowed-sinc interpolation 5

4 OFDM signal and channel reconstruction **5**

4.1 QPSK reference symbol 5

4.2 Frequency-domain channel estimate 6

4.3 Delay search and noise-estimation regions 6

5 Noise model and SNR derivation **7**

5.1 Complex AWGN model 7

5.2 Noise variance estimation from the reconstructed tail 8

6 Adaptive threshold from the target PFA **8**

6.1 Cell-level false-alarm probability 8

6.2 Cell PFA is not profile-level PFA 9

7 Local-maximum detection and sub-sample delay refinement **9**
7.1 Peak selection rule 9
7.2 Parabolic interpolation in the logarithmic magnitude domain 10

8 Noise-free feature catalogue and attribution logic **10**
8.1 Primary reference features 10
8.2 Expanded feature search and deterministic secondary features 11
8.3 One-to-one matching 11

9 Detection metrics and mathematical calculation **12**
9.1 Detection probability and recall 12
9.2 Precision 12
9.3 F1-score 12
9.4 Delay error metrics 13
9.5 Per-trial attribution counts 13

10 Monte Carlo procedure **14**

11 Recorded results **14**

12 Discussion of the results **16**
12.1 Effect of SNR 16
12.2 Effect of target PFA 16
12.3 Interpretation of the attribution results 16
12.4 Delay accuracy 17

13 Assumptions and limitations **17**

14 Conclusion **17**

A Generated figures **18**

1 Experiment objective and configuration

Experiment 17C investigates peak detection and false-alarm attribution when the simulated multipath channel is present. Unlike a noise-only detector test, the received OFDM symbol contains the channel response and additive complex Gaussian noise. The detector estimates the noise variance from a nominally noise-only tail of the reconstructed impulse response, calculates an adaptive magnitude threshold, detects local maxima, and matches detected delays against a noise-free reference catalogue.

The principal simulation parameters taken from the Python implementation are listed in Table 1.

Table 1: Experiment 17C configuration.

Parameter	Value
Carrier frequency, f_c	5.9 GHz
Sampling frequency, f_s	100 MHz
Sampling interval, $\Delta t = 1/f_s$	10 ns
OFDM FFT size, N_{FFT}	1024
Cyclic-prefix length	128 samples
Discrete channel maximum delay	600 ns
Sinc interpolation kernel radius	16
Delay search interval	0 ns to 400 ns
Ray adjacent-gap grouping threshold	10 ns
Primary matching tolerance	10 ns
Primary reference profile floor	−45 dB
Expanded local-maximum search floor	−60 dB
Secondary-feature matching tolerance	3 ns
SNR values	$-5, 0, 5, 10, 15, 20$ dB
Target cell PFA values	$10^{-2}, 10^{-3}, 10^{-4}$
Monte Carlo trials per SNR/PFA pair	1000
Random-number seed	2031

2 Multipath channel model

2.1 Continuous-time representation

The Sionna RT propagation solver returns a set of valid propagation paths. The equivalent baseband channel impulse response is represented as

$$h(\tau) = \sum_{\ell=1}^{L} \alpha_\ell \delta(\tau - \tau_\ell), \tag{1}$$

where L is the number of valid ray entries, α_ℓ is the complex path coefficient, τ_ℓ is the path delay, and $\delta(\cdot)$ is the Dirac delta function.

The path power associated with ray ℓ is

$$P_\ell = |\alpha_\ell|^2. \tag{2}$$

The complex coefficient can be written as

$$\alpha_\ell = a_{\ell,\mathrm{R}} + j a_{\ell,\mathrm{I}}, \tag{3}$$

where $a_{\ell,\mathrm{R}}$ and $a_{\ell,\mathrm{I}}$ are its real and imaginary components.

For this scene, the solver produced 12 valid ray entries. These are individual propagation contributions; they should not automatically be interpreted as 12 separately resolvable peaks in the sampled delay profile.

2.2 Ray grouping

The code sorts ray delays in ascending order and groups adjacent rays when their delay difference is no greater than the grouping threshold $\Delta_{\mathrm{group}} = 10\,\mathrm{ns}$. Thus, for two adjacent sorted rays,

$$\tau_{\ell+1} - \tau_\ell \le \Delta_{\mathrm{group}} \quad \Longrightarrow \quad \ell, \ell+1 \text{ belong to the same group.} \tag{4}$$

This is an adjacent-gap clustering rule. It does not require every pair of rays in a group to be within 10 ns; a chain of adjacent gaps can form one group.

For group g, the code calculates a power-weighted representative delay:

$$\boxed{\bar{\tau}_g = \frac{\sum\limits_{\ell \in G_g} |\alpha_\ell|^2 \tau_\ell}{\sum\limits_{\ell \in G_g} |\alpha_\ell|^2}} \tag{5}$$

provided that the sum of the ray powers is non-zero. If it is zero, the implementation uses the arithmetic mean of the group delays.

For example, the first group contains the two ray delays

$$\tau_1 = 133.425\,628\,7\,\mathrm{ns},$$
$$\tau_2 = 134.111\,709\,6\,\mathrm{ns}.$$

Its representative delay is therefore

$$\bar{\tau}_1 = \frac{|\alpha_1|^2(133.4256287) + |\alpha_2|^2(134.1117096)}{|\alpha_1|^2 + |\alpha_2|^2} = 133.598\,\mathrm{ns}, \tag{6}$$

using the path coefficients returned by Sionna RT. The coefficient powers are the weights; this is not necessarily the same as the unweighted mean.

Table 2: Ray-group ground truth obtained from the Experiment 17C run.

Group	Number of ray entries	Power-weighted delay (ns)
1	2	133.598
2	2	147.063
3	4	180.489
4	2	227.935
5	2	282.838

The five groups are used as propagation-ray support information. The subsequent detection reference, however, is formed from local maxima of the combined noise-free channel profile, not by simply using the five weighted group delays as the detector's reference delays.

3 Discrete-time channel construction

3.1 Sampling interval and delay grid

The sampling interval is

$$\Delta t = \frac{1}{f_s} = \frac{1}{100 \times 10^6} = 10\,\text{ns}. \tag{7}$$

A continuous path delay τ_ℓ maps to the fractional sample index

$$q_\ell = \frac{\tau_\ell}{\Delta t} = \tau_\ell f_s. \tag{8}$$

Since a ray delay is not generally an integer multiple of 10 ns, rounding every path to the nearest sample would introduce quantisation error. The implementation instead distributes each ray coefficient over nearby discrete samples using a windowed-sinc interpolation kernel.

3.2 Windowed-sinc interpolation

For integer sample index n, define the fractional offset

$$u_{\ell,n} = n - q_\ell. \tag{9}$$

The interpolation weight used by the script is

$$w_\ell[n] = \text{sinc}(u_{\ell,n}) \,\text{sinc}\left(\frac{u_{\ell,n}}{K+1}\right), \tag{10}$$

where $K = 16$ and the NumPy convention is $\text{sinc}(x) = \sin(\pi x)/(\pi x)$. Only samples satisfying

$$|u_{\ell,n}| < K + 1 \tag{11}$$

and lying inside the allocated channel array are retained. The discrete channel is accumulated as

$$\boxed{h[n] = \sum_{\ell=1}^{L} \alpha_\ell w_\ell[n].} \tag{12}$$

The maximum delay is 600 ns; at 10 ns per sample, the code allocates

$$N_h = \left\lceil \frac{600\,\text{ns}}{10\,\text{ns}} \right\rceil = 60 \tag{13}$$

samples. The interpolation kernel is truncated at the channel-array boundaries.

4 OFDM signal and channel reconstruction

4.1 QPSK reference symbol

The implementation creates one QPSK value on each of the 1024 subcarriers:

$$X[k] = \frac{(1 - 2b_{I,k}) + j(1 - 2b_{Q,k})}{\sqrt{2}}, \quad b_{I,k}, b_{Q,k} \in \{0, 1\}. \tag{14}$$

Consequently,

$$|X[k]|^2 = 1 \tag{15}$$

for every subcarrier. The time-domain OFDM symbol is

$$x[n] = \frac{1}{N_{\mathrm{FFT}}} \sum_{k=0}^{N_{\mathrm{FFT}}-1} X[k] e^{j2\pi kn/N_{\mathrm{FFT}}}, \tag{16}$$

using the standard inverse-DFT normalisation of NumPy.

The cyclic prefix is formed by copying the final 128 samples of $x[n]$ to the beginning of the symbol. The channel output is generated by linear convolution:

$$y_{\mathrm{full}}[n] = (x_{\mathrm{CP}} * h)[n]. \tag{17}$$

The useful N_{FFT} samples are selected after discarding the cyclic-prefix interval:

$$y_{\mathrm{clean}}[n] = y_{\mathrm{full}}[n + N_{\mathrm{CP}}], \quad 0 \le n < N_{\mathrm{FFT}}. \tag{18}$$

The script checks that the cyclic prefix is long enough for the allocated discrete channel. Here $N_{\mathrm{CP}} = 128$ and $N_h - 1 = 59$, so the check passes.

4.2 Frequency-domain channel estimate

The frequency-domain received symbol is

$$Y[k] = \mathrm{FFT}\{y_{\mathrm{clean}}[n]\}. \tag{19}$$

The channel estimate is obtained by per-subcarrier division:

$$\boxed{\hat{H}[k] = \frac{Y[k]}{X[k]}.} \tag{20}$$

Because the transmitted QPSK symbols have unit magnitude, no subcarrier division by zero or small-magnitude pilot issue occurs. The estimated discrete impulse response is

$$\hat{h}[n] = \mathrm{IFFT}\{\hat{H}[k]\}. \tag{21}$$

In the noise-free case, this procedure reconstructs the discrete channel used to generate the OFDM symbol, subject to the implementation's finite FFT, cyclic-prefix, and channel-window assumptions.

4.3 Delay search and noise-estimation regions

The detector searches from zero to 400 ns. With a 10 ns sampling interval, the code sets

$$n_{\mathrm{search,end}} = \left\lfloor \frac{400}{10} \right\rfloor + 1 = 41. \tag{22}$$

Thus, the searched sample indices are 0 through 40, corresponding to 0–400 ns.

The noise-estimation region begins at

$$n_{\mathrm{noise,start}} = \left\lceil \frac{600\,\mathrm{ns}}{10\,\mathrm{ns}} \right\rceil = 60. \tag{23}$$

It therefore uses indices 60 through 1023, giving

$$N_{\mathrm{noise}} = 1024 - 60 = 964 \tag{24}$$

samples. The code verifies that the search and noise-estimation regions do not overlap.

5 Noise model and SNR derivation

5.1 Complex AWGN model

The noisy useful OFDM symbol is modelled as

$$y_{\text{noisy}}[n] = y_{\text{clean}}[n] + v[n], \tag{25}$$

where

$$v[n] = \sqrt{\frac{\sigma_v^2}{2}}\,(v_I[n] + j v_Q[n])\,, \tag{26}$$

and $v_I[n], v_Q[n] \sim \mathcal{N}(0,1)$ are independent. Hence

$$\mathbb{E}\{|v[n]|^2\} = \sigma_v^2. \tag{27}$$

The average received signal power measured by the script is

$$P_s = \frac{1}{N_{\text{FFT}}} \sum_{n=0}^{N_{\text{FFT}}-1} |y_{\text{clean}}[n]|^2. \tag{28}$$

The measured value for the recorded run was

$$\boxed{P_s = 1.258177 \times 10^{-11}.} \tag{29}$$

The linear SNR is defined by

$$\text{SNR}_{\text{lin}} = \frac{P_s}{\sigma_v^2}. \tag{30}$$

Since $\text{SNR}_{\text{dB}} = 10\log_{10}(\text{SNR}_{\text{lin}})$, the noise variance used by the code is

$$\boxed{\sigma_v^2 = P_s\, 10^{-\text{SNR}_{\text{dB}}/10}.} \tag{31}$$

For example, at 10 dB:

$$\sigma_v^2 = 1.258177 \times 10^{-11} \times 10^{-10/10} \tag{32}$$

$$= 1.258177 \times 10^{-11} \times 0.1 \tag{33}$$

$$= \boxed{1.258177 \times 10^{-12}.} \tag{34}$$

The corresponding theoretical noise variances for all tested SNR values are given in Table 3.

Table 3: Noise variance calculated from the measured signal power.

SNR (dB)	σ_v^2
-5	3.978705×10^{-11}
0	1.258177×10^{-11}
5	3.978705×10^{-12}
10	1.258177×10^{-12}
15	3.978705×10^{-13}
20	1.258177×10^{-13}

5.2 Noise variance estimation from the reconstructed tail

After channel estimation and inverse FFT, the code estimates noise variance from the tail of $\hat{h}[n]$. For circular complex Gaussian noise $z \sim \mathcal{CN}(0, \sigma^2)$, the squared magnitude has an exponential distribution:

$$f_{|z|^2}(p) = \frac{1}{\sigma^2} \exp\left(-\frac{p}{\sigma^2}\right), \quad p \geq 0. \tag{35}$$

Its cumulative distribution is

$$F_{|z|^2}(p) = 1 - \exp\left(-\frac{p}{\sigma^2}\right). \tag{36}$$

The median m satisfies $F(m) = 0.5$, so

$$0.5 = 1 - \exp(-m/\sigma^2), \tag{37}$$

$$\exp(-m/\sigma^2) = 0.5, \tag{38}$$

$$m = \sigma^2 \ln 2. \tag{39}$$

Therefore, a median-based estimator is

$$\boxed{\hat{\sigma}^{\,2} = \frac{\text{median}(|\hat{h}[n]|^2)}{\ln 2}.} \tag{40}$$

This is exactly the estimator implemented in the script. A lower numerical bound of 10^{-30} is applied to prevent a zero estimate.

The estimate is taken from indices $n \geq 60$. It is an estimate of the noise variance in the reconstructed impulse-response tail, rather than a direct sample-by-sample estimate from the original received time-domain noise. The interpretation relies on the tail being effectively free of significant channel energy.

6 Adaptive threshold from the target PFA

6.1 Cell-level false-alarm probability

Let z be a noise-only complex Gaussian sample with $z \sim \mathcal{CN}(0, \sigma^2)$. The magnitude $r = |z|$ has a Rayleigh distribution, and its tail probability is

$$\Pr(|z| > T) = \exp\left(-\frac{T^2}{\sigma^2}\right). \tag{41}$$

Setting this probability equal to a target cell-level PFA, P_{FA}, gives

$$P_{\text{FA}} = \exp\left(-\frac{T^2}{\sigma^2}\right), \tag{42}$$

$$\ln(P_{\text{FA}}) = -\frac{T^2}{\sigma^2}, \tag{43}$$

$$\boxed{T = \sqrt{-\sigma^2 \ln(P_{\text{FA}})}.} \tag{44}$$

The script substitutes the estimated variance $\hat{\sigma}^{\,2}$:

$$\boxed{T = \sqrt{-\hat{\sigma}^{\,2} \ln(P_{\text{FA}})}.} \tag{45}$$

This is a magnitude threshold, not a power threshold. The detector compares $|\hat{h}[n]|$ directly with T.

For example, at a target cell PFA of 10^{-3},

$$T = \sqrt{-\hat{\sigma}^2 \ln(10^{-3})} = \sqrt{6.907755\,\hat{\sigma}^2}. \tag{46}$$

If the variance estimate equals the theoretical variance at 10 dB, then

$$T = \sqrt{6.907755(1.258177 \times 10^{-12})} \tag{47}$$

$$\approx \boxed{2.9481 \times 10^{-6}}. \tag{48}$$

This is an illustrative theoretical substitution; the actual threshold in each Monte Carlo trial is calculated from that trial's estimated $\hat{\sigma}^2$.

Table 4: Illustrative theoretical magnitude thresholds using the nominal noise variance and measured P_s. These are not the per-trial estimated thresholds stored by the program.

SNR (dB)	$P_{\text{FA}} = 10^{-2}$	$P_{\text{FA}} = 10^{-3}$	$P_{\text{FA}} = 10^{-4}$
-5	$1.353\,61 \cdot 10^{-5}$	$1.657\,83 \cdot 10^{-5}$	$1.914\,29 \cdot 10^{-5}$
0	$7.611\,91 \cdot 10^{-6}$	$9.322\,65 \cdot 10^{-6}$	$1.076\,49 \cdot 10^{-5}$
5	$4.280\,49 \cdot 10^{-6}$	$5.242\,51 \cdot 10^{-6}$	$6.053\,53 \cdot 10^{-6}$
10	$2.407\,10 \cdot 10^{-6}$	$2.948\,08 \cdot 10^{-6}$	$3.404\,15 \cdot 10^{-6}$
15	$1.353\,61 \cdot 10^{-6}$	$1.657\,83 \cdot 10^{-6}$	$1.914\,29 \cdot 10^{-6}$
20	$7.611\,91 \cdot 10^{-7}$	$9.322\,65 \cdot 10^{-7}$	$1.076\,49 \cdot 10^{-6}$

6.2 Cell PFA is not profile-level PFA

The target P_{FA} in Equation (44) is the probability that one noise-only complex sample exceeds the magnitude threshold. If M independent noise-only cells are tested, the probability of at least one cell exceedance is

$$P_{\text{FA,profile}} = 1 - (1 - P_{\text{FA}})^M. \tag{49}$$

For small P_{FA}, this is approximately MP_{FA} when MP_{FA} is sufficiently small. This expression is an idealised independent-cell calculation. The actual experiment detects local maxima in a reconstructed profile, whose samples and peak events need not be independent. Therefore the number of local maxima per trial must be measured empirically; it must not be equated directly with the cell-level PFA.

7 Local-maximum detection and sub-sample delay refinement

7.1 Peak selection rule

Let $a[n] = |\hat{h}[n]|$ denote the reconstructed magnitude profile. A sample n is accepted as a detected local maximum if

$$\boxed{a[n] \geq T, \qquad a[n] > a[n-1], \qquad a[n] \geq a[n+1].} \tag{50}$$

The implementation searches only the specified delay interval and excludes the two boundary samples from local-maximum testing. The asymmetric strict and non-strict comparisons are intentional: the left neighbour must be strictly smaller, while the right neighbour may be equal.

7.2 Parabolic interpolation in the logarithmic magnitude domain

The integer peak index provides a delay estimate with a nominal grid spacing of 10 ns. The code refines this location using a three-point parabolic interpolation on the logarithmic magnitude values.

Define

$$L_{-1} = 20\log_{10}(a[n-1]), \tag{51}$$
$$L_0 = 20\log_{10}(a[n]), \tag{52}$$
$$L_{+1} = 20\log_{10}(a[n+1]). \tag{53}$$

The estimated fractional offset from the centre sample is

$$\boxed{\delta = \frac{1}{2}\frac{L_{-1} - L_{+1}}{L_{-1} - 2L_0 + L_{+1}}.} \tag{54}$$

The code clips δ to the interval $[-0.5, 0.5]$. The refined sample position and delay are

$$\hat{n} = n + \delta, \tag{55}$$
$$\boxed{\hat{\tau} = (n+\delta)\Delta t.} \tag{56}$$

When the denominator is numerically close to zero, the code uses the integer peak index instead. The interpolation improves the numerical location estimate, but does not change the underlying 10 ns sample spacing or guarantee that two physically close paths are resolvable.

8 Noise-free feature catalogue and attribution logic

8.1 Primary reference features

The clean reconstructed profile is

$$a_0[n] = |\hat{h}_{\text{clean}}[n]|. \tag{57}$$

The primary reference threshold is set relative to the maximum magnitude within the search interval:

$$T_{\text{primary}} = \max_{n\in\mathcal{S}} a_0[n]\, 10^{-45/20}, \tag{58}$$

where $\mathcal{S}$ denotes the search samples. The use of -45 dB here is a magnitude ratio expressed in decibels, hence the factor 20 in the exponent.

The relative magnitude profile is

$$A_{\text{dB}}[n] = 20\log_{10}\left(\frac{a_0[n]}{\max_{m\in\mathcal{S}} a_0[m]}\right). \tag{59}$$

The noise-free local maxima above the -45 dB floor form the primary profile reference catalogue.

The recorded noise-free reference peaks are shown in Table 5.

Table 5: Noise-free composite-profile reference peaks.

Peak	Refined delay (ns)	Relative magnitude (dB)
1	131.151	-4.27
2	149.601	0.00
3	176.494	-16.01
4	230.030	-24.81
5	281.621	-22.94

These peak delays differ from the weighted ray-group delays in Table 2. The ray-group delays summarise the individual Sionna RT rays. The profile peak delays are local maxima of the combined, interpolated, discrete channel response. Complex-valued ray contributions combine before the magnitude profile is formed, so their peak positions need not equal the power-weighted average delays.

8.2 Expanded feature search and deterministic secondary features

The script repeats local-maximum detection on the clean profile using a lower −60 dB magnitude floor:

$$T_{\text{expanded}} = \max_{n \in \mathcal{S}} a_0[n]\, 10^{-60/20}. \tag{60}$$

Each expanded local maximum is compared with the nearest ray-group delay. If its distance from a ray group is no greater than the code's 10 ns primary matching tolerance, it is labelled `ray-supported-primary`. Otherwise it is labelled `deterministic-secondary-feature`.

The implementation then forms the primary catalogue by retaining the features above the −45 dB floor. Features found only through the expanded search are retained for the secondary-feature attribution step. In the recorded run, the expanded search found five local maxima, all of which were ray-group supported; no deterministic secondary feature was identified by this operational rule.

This result means that no additional stable local maximum was found above the selected −60 dB floor in the searched noise-free profile. It does not prove that the physical propagation environment contains no weaker paths or that no unresolved channel structure exists.

8.3 One-to-one matching

For a reference delay τ_i and detected delay $\hat{\tau}_j$, define the signed error

$$e_{ij} = \hat{\tau}_j - \tau_i. \tag{61}$$

A candidate match is permitted if

$$|e_{ij}| \le \epsilon, \qquad \epsilon = 10\,\text{ns}. \tag{62}$$

The code creates all candidate pairs satisfying this condition, sorts them by absolute error, and greedily accepts a pair only if neither the reference nor the detection has already been used. This enforces one-to-one matching.

Let N_{ref} be the number of primary reference peaks and N_{TP} the number matched in a trial. Then

$$N_{\text{FN}} = N_{\text{ref}} - N_{\text{TP}}, \tag{63}$$

$$N_{\text{unmatched}} = N_{\text{det}} - N_{\text{matched,det}}. \tag{64}$$

The primary reference matching is performed before secondary-feature matching. An unmatched detection is subsequently compared with deterministic secondary features using the tighter 3 ns tolerance. Any detection still unmatched is labelled `noise-or-unmodelled-channel-detection`.

9 Detection metrics and mathematical calculation

9.1 Detection probability and recall

Across N_{trials} trials, the total number of reference opportunities is

$$N_{\text{opportunities}} = N_{\text{trials}} N_{\text{ref}}. \tag{65}$$

The profile detection probability calculated by the script is

$$\boxed{P_D = \frac{N_{\text{TP}}}{N_{\text{trials}} N_{\text{ref}}} \times 100\%.} \tag{66}$$

The script's recall is

$$\text{Recall} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}}. \tag{67}$$

Since $N_{\text{TP}} + N_{\text{FN}}$ equals the total number of reference opportunities, the reported detection probability and recall are numerically identical.

9.2 Precision

The script defines the number of unmatched detections after primary matching as the false-positive count for primary-profile precision:

$$N_{\text{FP}} = N_{\text{unmatched}}. \tag{68}$$

Therefore,

$$\boxed{\text{Precision} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{unmatched}}}.} \tag{69}$$

An important implementation detail is that this denominator includes detections later attributed to deterministic secondary features. Thus, secondary detections remain unmatched with respect to the primary catalogue and reduce primary-profile precision, even though the attribution analysis can identify them separately.

9.3 F1-score

The F1-score is the harmonic mean of precision and recall:

$$\boxed{F_1 = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.} \tag{70}$$

For illustration, the recorded result at 20 dB and target PFA 10^{-3} was approximately

$$\text{Recall} = 1.0000,$$
$$\text{Precision} = 0.8409.$$

Substitution gives

$$F_1 = 2\frac{1.0000 \times 0.8409}{1.0000 + 0.8409} \approx 0.9139,$$

consistent with the reported rounded F1-score of approximately 0.914.

9.4 Delay error metrics

For every primary reference-to-detection match, the signed delay error is

$$e_i = \hat{\tau}_i - \tau_i. \tag{71}$$

For N_m matched detections, the mean absolute error is

$$\boxed{\mathrm{MAE} = \frac{1}{N_m} \sum_{i=1}^{N_m} |e_i|.} \tag{72}$$

The root mean square error is

$$\boxed{\mathrm{RMSE} = \sqrt{\frac{1}{N_m} \sum_{i=1}^{N_m} e_i^2}.} \tag{73}$$

The signed bias is

$$\boxed{\mathrm{Bias} = \frac{1}{N_m} \sum_{i=1}^{N_m} e_i.} \tag{74}$$

The script reports MAE and RMSE. These errors are calculated only for primary one-to-one matches; unmatched detections do not enter the delay error calculation.

9.5 Per-trial attribution counts

For each trial, the code partitions detected peaks into three operational categories:

$$\boxed{N_{\mathrm{det}} = N_{\mathrm{primary}} + N_{\mathrm{secondary}} + N_{\mathrm{noise/unmodelled}}.} \tag{75}$$

Here:

- N_{primary} is the number of detections matched to the primary noise-free profile catalogue.
- $N_{\mathrm{secondary}}$ is the number of remaining detections matched to a deterministic secondary feature using the 3 ns tolerance.
- $N_{\mathrm{noise/unmodelled}}$ is the residual number of detections after both matching stages.

The reported mean count per trial is calculated by dividing the accumulated count by the number of Monte Carlo trials. For example,

$$\overline{N}_{\mathrm{noise/unmodelled}} = \frac{\sum_{r=1}^{N_{\mathrm{trials}}} N_{\mathrm{noise/unmodelled},r}}{N_{\mathrm{trials}}}. \tag{76}$$

This final category must not be described as confirmed noise false alarms: it can also contain unresolved or unmodelled channel features, and detections that fail the specified matching tolerances.

The ray-supported detection count is the number of primary matched detections whose corresponding reference features are labelled `ray-supported-primary`. Its group-level recall is calculated as

$$R_{\mathrm{ray}} = \frac{N_{\mathrm{ray-supported}}}{N_{\mathrm{trials}} N_{\mathrm{ray-groups}}} \times 100\%. \tag{77}$$

This is distinct from profile detection probability: the denominator is based on the number of ray groups rather than the number of composite profile peaks.

10 Monte Carlo procedure

For every SNR and target PFA combination, the script performs 1000 trials. For each trial:

1. Generate complex AWGN with the variance from Equation (31).
2. Add the noise to the clean received OFDM symbol.
3. Estimate the frequency response by dividing the received FFT by the known QPSK symbol.
4. Apply the inverse FFT to obtain the noisy channel estimate.
5. Estimate noise variance from the tail using Equation (40).
6. Calculate the adaptive magnitude threshold using Equation (44).
7. Detect local maxima in the 0–400 ns search interval.
8. Refine detected peak positions using Equation (54).
9. Match detections to primary reference peaks using the 10 ns one-to-one matching tolerance.
10. Match remaining detections to deterministic secondary features using the 3 ns tolerance.
11. Accumulate detection, attribution, and delay-error statistics.

The random generator is initialised with seed 2031. The same generator is used sequentially through the SNR/PFA loops, so the results are reproducible for the same software environment and execution order.

11 Recorded results

The following tables reproduce the Experiment 17C result values recorded for the completed run. Detection probability, precision, F1-score, and RMSE are reported for each SNR and target PFA setting. Counts per trial are average counts over the Monte Carlo trials.

Table 6: Profile detection probability (%).

SNR (dB)	$P_{\mathrm{FA}} = 10^{-2}$	$P_{\mathrm{FA}} = 10^{-3}$	$P_{\mathrm{FA}} = 10^{-4}$
-5	54.42	47.70	43.18
0	67.38	58.00	55.16
5	91.52	84.24	74.56
10	98.36	98.08	97.90
15	99.90	99.86	99.88
20	100.00	100.00	100.00

Table 7: Primary-profile precision (%).

SNR (dB)	$P_{\mathrm{FA}} = 10^{-2}$	$P_{\mathrm{FA}} = 10^{-3}$	$P_{\mathrm{FA}} = 10^{-4}$
-5	78.53	91.91	97.16
0	72.69	85.02	92.58
5	71.26	79.77	85.06
10	70.94	78.67	84.19
15	74.39	79.47	84.14
20	77.93	84.09	86.91

Table 8: F1-score.

SNR (dB)	$P_{\mathrm{FA}} = 10^{-2}$	$P_{\mathrm{FA}} = 10^{-3}$	$P_{\mathrm{FA}} = 10^{-4}$
-5	0.643	0.628	0.598
0	0.699	0.690	0.691
5	0.801	0.819	0.795
10	0.824	0.873	0.905
15	0.853	0.885	0.913
20	0.876	0.914	0.930

Table 9: Mean delay RMSE (ns).

SNR (dB)	$P_{\mathrm{FA}} = 10^{-2}$	$P_{\mathrm{FA}} = 10^{-3}$	$P_{\mathrm{FA}} = 10^{-4}$
-5	2.1811	1.6565	1.1789
0	2.2127	1.6558	1.3688
5	2.0475	1.8107	1.4843
10	1.2069	1.1742	1.1713
15	0.6077	0.6291	0.6336
20	0.3386	0.3376	0.3353

Table 10: Mean per-trial detection attribution counts.

SNR (dB)	Target PFA	Ray-supported	Secondary	Noise/unmodelled
-5	10^{-2}	2.721	0.000	0.744
-5	10^{-3}	2.385	0.000	0.210
-5	10^{-4}	2.159	0.000	0.063
0	10^{-2}	3.369	0.000	1.266
0	10^{-3}	2.900	0.000	0.511
0	10^{-4}	2.758	0.000	0.221
5	10^{-2}	4.576	0.000	1.846
5	10^{-3}	4.212	0.000	1.068
5	10^{-4}	3.728	0.000	0.655
10	10^{-2}	4.918	0.000	2.015
10	10^{-3}	4.904	0.000	1.330
10	10^{-4}	4.895	0.000	0.919
15	10^{-2}	4.995	0.000	1.720
15	10^{-3}	4.993	0.000	1.290
15	10^{-4}	4.994	0.000	0.941
20	10^{-2}	5.000	0.000	1.416
20	10^{-3}	5.000	0.000	0.946
20	10^{-4}	5.000	0.000	0.753

12 Discussion of the results

12.1 Effect of SNR

Detection probability generally increases with SNR because the channel features become more distinguishable from the noise floor. At 20 dB, all five primary reference features were detected in every trial for the three tested target PFA values in the recorded results. At lower SNR, weak features are more likely to fall below the adaptive threshold, increasing the number of missed reference features.

12.2 Effect of target PFA

A smaller target cell PFA produces a higher threshold through Equation (44). This generally reduces noise-driven detections, but it can also suppress weak channel features. The results show this trade-off particularly at low and intermediate SNR values: reducing the target PFA tends to improve precision while reducing detection probability.

The target PFA is therefore a detector operating parameter, not a direct guarantee of the final number of false local maxima per delay profile.

12.3 Interpretation of the attribution results

No detections were assigned to deterministic secondary features in the recorded result table. This follows from the noise-free expanded local maximum search: all five detected clean-profile local maxima were classified as ray-group-supported primary features. Consequently, the unmatched noisy detections were assigned to the residual noise/unmodelled category.

The label *noise/unmodelled* is deliberately conservative. The algorithm establishes that a

detection did not match either of the catalogued feature types under the specified tolerances. It does not establish the physical cause of that detection. A residual detection may arise from noise, unresolved channel structure, modelling limitations, interpolation effects, or a matching-tolerance limitation.

12.4 Delay accuracy

The RMSE decreases substantially as SNR increases. At 20 dB, the reported RMSE is approximately 0.34 ns, which is smaller than the 10 ns sample interval. This is possible because the detector applies three-point parabolic interpolation to estimate a fractional sample position. It should not be interpreted as evidence that the sampled waveform has a physical delay resolution of 0.34 ns; it is the empirical error of the interpolated peak estimator under this simulated profile and matching procedure.

13 Assumptions and limitations

The following limitations are important when interpreting the findings:

1. The propagation channel is generated by the Sionna RT street-canyon scene and the specified transmitter and receiver positions. The results are simulation-based, not an SDR or on-road hardware validation.
2. The channel is treated as static over the OFDM symbol and Monte Carlo trials. The experiment does not model Doppler evolution or time-varying vehicle motion.
3. The median noise estimator assumes that the selected tail is sufficiently free of channel energy and that the reconstructed noise samples are adequately represented by the assumed complex Gaussian model.
4. The threshold is designed from a cell-level Rayleigh-tail probability. The actual local-maximum false-alarm behaviour is affected by the search length, local-maximum rule, sample dependence, and peak refinement.
5. Ray grouping uses a fixed adjacent-gap threshold of 10 ns. Different grouping thresholds may produce different group catalogues.
6. Primary and secondary attribution depends on the chosen reference floors and matching tolerances. It is an operational classification, not a proof of the physical origin of each detection.
7. The five primary profile peaks are composite response maxima. They are not necessarily in one-to-one correspondence with individual rays.
8. The residual category must be reported as *noise/unmodelled-channel detections*, not as confirmed noise-only false alarms.

14 Conclusion

Experiment 17C establishes a reproducible simulation workflow for channel-present multipath peak detection and attribution. The mathematical chain begins with the Sionna RT multipath representation, groups closely spaced ray entries using a power-weighted delay, constructs a fractional delay discrete channel using a windowed-sinc kernel, and estimates the channel impulse

response from a known QPSK OFDM symbol. Complex Gaussian noise is added at controlled SNR levels, and a median-based estimator provides a per-trial noise variance estimate. An adaptive Rayleigh-tail magnitude threshold is then used to detect local maxima.

The detector's output is compared with a noise-free composite-profile catalogue using greedy one-to-one delay matching. Remaining detections are checked against the deterministic-secondary feature catalogue and then classified as noise/unmodelled if no match is found. Across the recorded results, detection probability improves with SNR, while lower target PFA values generally trade detection probability for improved precision. Sub-sample parabolic interpolation yields delay errors below one sample interval at high SNR in this particular simulation.

The main scientific qualification is that the experiment's attribution scheme identifies detections unsupported by its noise-free feature catalogue; it does not prove that all such detections are caused by noise. The results should therefore be presented as simulation-based detector performance and feature attribution, with hardware validation and time-varying V2X channel evaluation remaining separate future steps.

A Generated figures

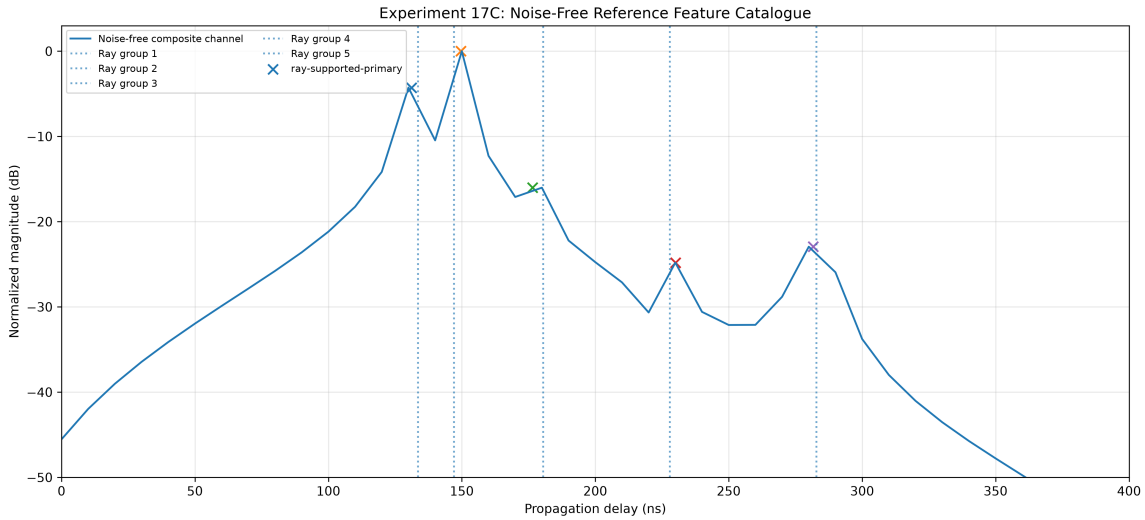


Figure 1: Noise-free composite-profile feature catalogue and ray-group delays.

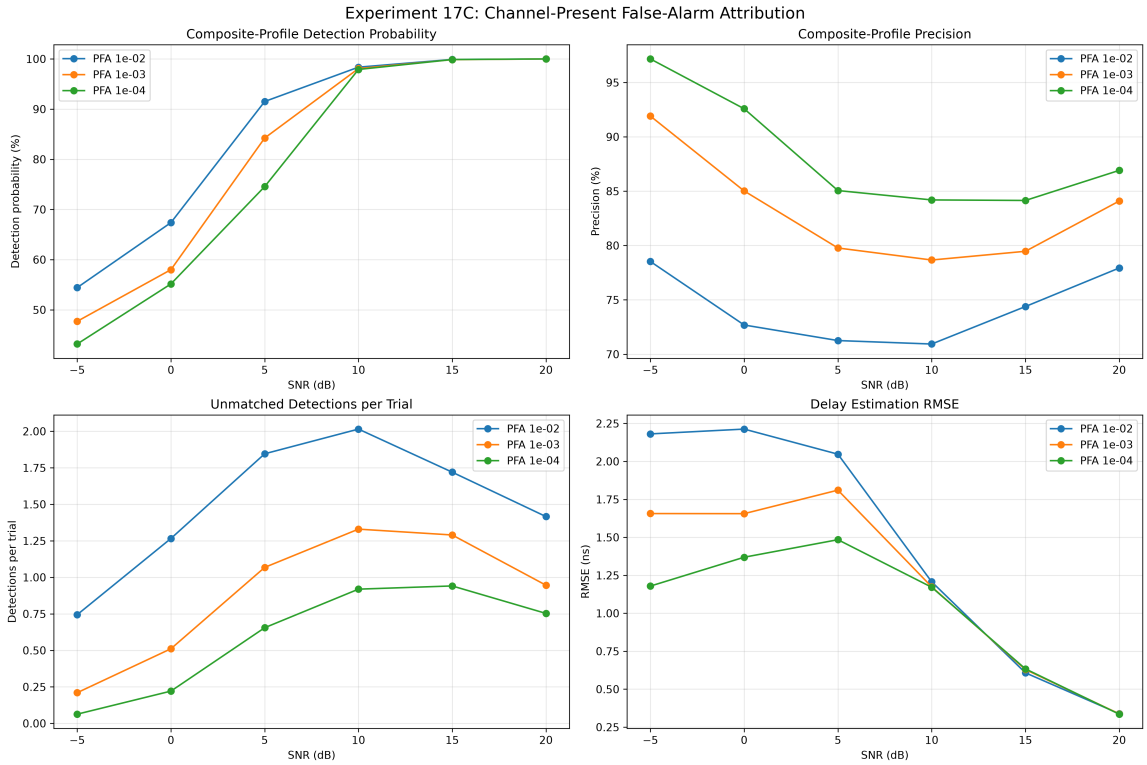


Figure 2: Experiment 17C Monte Carlo detection and attribution metrics.